EPM213s - Electrical Measurements

Lab report – 5

Exp 5: PMMC inspection and construction

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PMMC inspection and construction

1. Objective

To study the construction and working principle of a Permanent Magnet Moving Coil Instrument (PMMC Instrument).

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2. Construction of a PMMC Instrument

- **Permanent Magnet:**
  A strong and stable magnetic field is created using a permanent magnet, often shaped like a horseshoe to ensure a radial and uniform field.
- **Moving Coil:**
  A fine copper wire is wound into a coil around a lightweight, non-magnetic aluminum former, which allows movement when current flows.
- **Soft Iron Core:**
  Positioned within the coil to enhance the magnetic field's strength and ensure uniformity.
- **Control Springs:**
  These springs not only generate the controlling torque but also act as electrical connectors to the coil.
- **Pointer and Scale:**
  The moving coil is attached to a spindle connected to a pointer, which moves across a calibrated scale to display the measurement.
- **Damping Mechanism:**
  Damping is achieved through eddy currents induced in the

aluminum former, helping the pointer reach a steady position quickly without oscillation.

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3. Types of Torques in PMMC Instruments

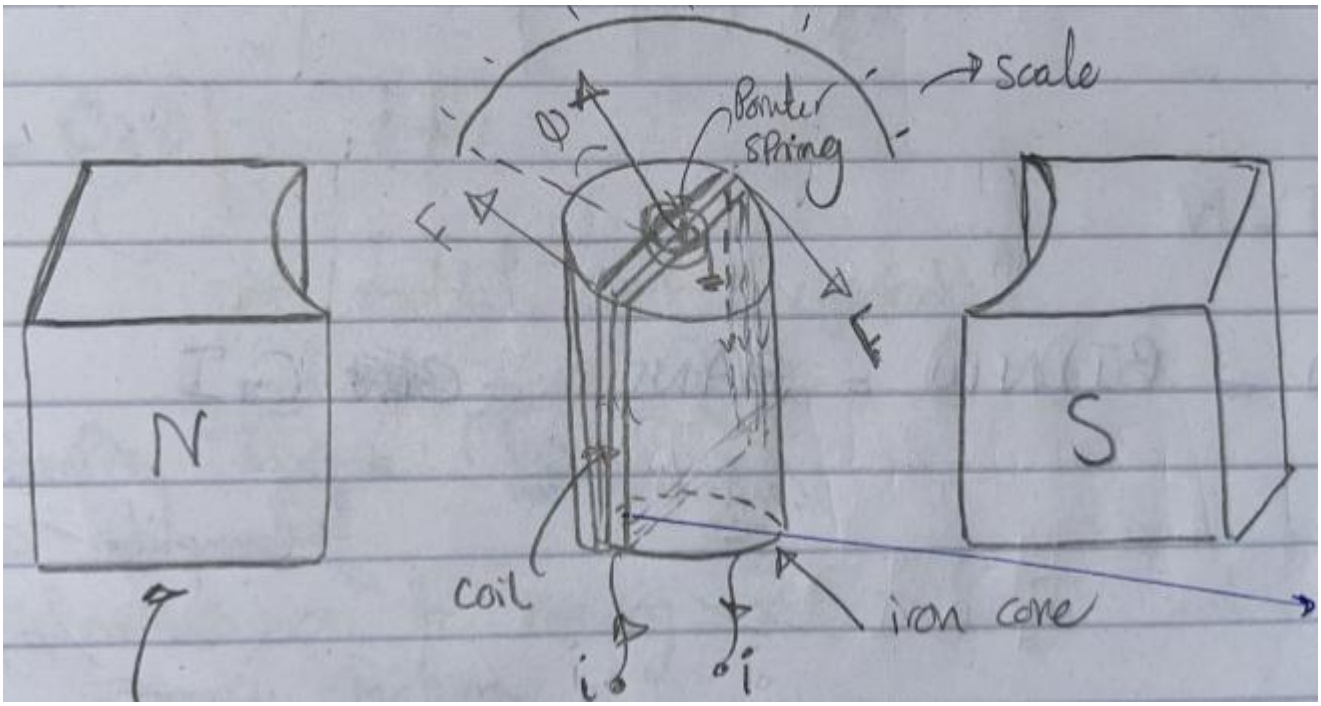
- **Deflecting Torque**
- **Controlling Torque**
- **Damping Torque**

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4. Working Principle of a PMMC Instrument

- **When a direct current (DC) flows through the coil, it experiences a force due to interaction with the magnetic field of the permanent magnet.**
- **This results in a deflecting torque (T = BINA), which causes the coil and the pointer to rotate.**
- **As the coil turns, controlling torque from the springs builds up and counteracts the deflecting torque.**
- **The coil settles at a position where both torques balance each other.**
- **The amount of deflection is directly proportional to the current flowing through the coil, resulting in a linear scale for measurements.**

5. Draw of PMMC:



6. Photos of PMMC:



